

theories Eq. (29.3.12) becomes instead

$$b = -\frac{1}{16\pi^2} (3C_1 - C_2) . \quad (29.3.13)$$

The solution of Eq. (29.3.11) for the running gauge coupling is then

$$g_\lambda^{-2} = g^{-2} + \frac{3C_1 - C_2}{8\pi^2} \ln \left(\frac{\lambda}{K} \right) , \quad (29.3.14)$$

where K is an ultraviolet cut-off, introduced to give meaning to the otherwise ultraviolet-divergent bare gauge coupling g .

To summarize the results so far, setting $T = \tau$ and $Y = 1$, the Wilsonian effective Lagrangian for $C_1 > C_2$ takes the form

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, \tau, \tau^*, \mathcal{D} \cdots) \right]_D + 2 \operatorname{Re} \left[\frac{\tau_\lambda}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}} \\ & + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} + \exp \left(\frac{2i\pi\tau_\lambda}{C_1 - C_2} \right) [v_\lambda(\Phi)]_{\mathcal{F}} , \end{aligned} \quad (29.3.15)$$

where

$$\tau_\lambda = \frac{4\pi i}{g_\lambda^2} + \frac{\theta}{2\pi} . \quad (29.3.16)$$

We have been able to replace τ with τ_λ in the exponential in the last term in Eq. (29.3.15), because the difference is a constant times $\ln \lambda$, which yields a power of λ that can be absorbed into the definition of v_λ .

Non-perturbative effects have now been isolated in the last term in Eq. (29.3.15). This term can be generated by instantons of winding number $\nu > 0$ if $C_1 - C_2 = 1/\nu$. (In general $C_1 - C_2$ is a rational number. For the generalized supersymmetric version of quantum chromodynamics, $C_1 - C_2 = N_c - N_f$ is an integer, so the condition $C_1 - C_2 = 1/\nu$ requires that $N_c = N_f - 1$, and then only $\nu = 1$ instantons contribute. Detailed calculations⁵ in this model show that instantons actually do make such contributions.) Whether or not it is instantons that generate the non-perturbative contribution $v_\lambda(\Phi)$, we can determine its form by considering the non-anomalous symmetries of the theory. Since this function is independent of Y , it can be evaluated as if $Y = 0$, so it shares all the non-anomalous symmetries of the first term in Eq. (29.3.1). These include the gauge symmetry itself and a global symmetry under $\prod_d SU(n(d))$, where d labels the different irreducible representations of the gauge group furnished by the left-chiral superfields, and $n(d)$ is the number of times representation d occurs. (For instance, in generalized supersymmetric quantum chromodynamics d takes two values, labelling the N_c and $\bar{N}_c$ representations of $SU(N_c)$, and $n(N_c) = n(\bar{N}_c) = N_f$.) Let us label the Φ s as $\Phi_{ai}^{(d)}$, where a is a gauge index, and i is a ‘flavor’ index, labelling